

ActInf GuestStream 082.3 ~ "The Projective Wave Theory of Consciousness"

GuestStream | Worden, Rudrauf, Williford | 1:22:39

Hello and welcome everyone. It's July 22, 2024. This is Active Inference Guest Stream 82.3, the Projective Wave Theory of Consciousness. Robert Worden will give a presentation and we have several others to join and comment and give their insights. So thank you everyone for joining and Robert please begin the presentation. And also Robert, close the YouTube on your side.

All good.

It's coming through my loudspeaker which is a problem.

Yeah, it's all good. Like, all good.

Okay, so this is a talk about the Projective Wave Theory of Consciousness and there's already a paper about it on

the archive this month so you can look at the paper if you like. But let me start by describing roughly the roadmap for the talk.

The Projective Wave Theory is very closely related to the Projective Consciousness Model, PCM, and we're fortunate to have three of the authors of that model on the call today.

But I'm going to give this talk in four different parts. The first part is talking about data and the idea is that any theory of consciousness has to fit the data, has to fit some data in order for us to believe it.

And so the question is, what kinds of data is there about consciousness that the theory has to fit?

And I shall be arguing that the main kind of data we have is spatial consciousness.

So that's part one of the talk. Part two, I shall talk about the Projective Consciousness Model, PCM.

And to my mind, the big virtue of the PCM is that it does predict, account for just that, the spatial nature of consciousness.

And in that, it's a great deal, it does a great deal better than many neural theories of consciousness.

And I shall probably pause at that point after parts one and two and have discussion about the first two points there.

But before that, I think the PCM has great strengths, but I think in common with all neural theories of consciousness, it has three problems.

And I shall talk about what those problems are.

But I shall emphasize that those problems are not specific to the PCM.

If you buy into the classical neural model of the brain that neurons do everything, then whatever your theory of consciousness, I think you're going to have these problems.

And so in order to get out of those problems, we have to do something that's not in the classical neural model that assumes that neurons in the brain do everything.

So we have to add something to the PCM.

And the something we have to add is there in three.

The assumption we're going to add to it is quite a big one.

It's an assumption that there is a wave in the brain which stores 3D spatial information.

In other words, neurons don't do everything.

And then part four of the talk, I shall bring those two and three together to a projective wave theory of consciousness and talk a little about firstly how that solves the neural problems.

But secondly, how it fits the data and some more general considerations about it.

So that is the route map of the talk.

So we start off with what are the reasons to believe some theory, whether it's theory of consciousness or anything else.

Why do we believe it?

Basically, we believe a theory because it fits data.

And there's a complexity measure here that needs to fit more data, which you can measure in bits, if you like, than the complexity of the assumptions you put into it.

If you just put in a huge number of assumptions and fit a small amount of data, that's no use.

So there's a kind of Bayesian balance between the assumptions you put into a theory and the data it fits.

And I've illustrated the Bayesian balance pair of scales here.

And I've measured the complexity of the theory and the amount of data it tests by a so-called zip test, which simply says you describe these things in text files, you zip them up and you see how big they are.

So established theories in science, like the theories of physics or chemistry, have a hugely positive Bayesian balance.

They have quite few assumptions and they account for huge amounts of data.

So how can we get to that point of the theory of consciousness?

How can a theory of consciousness produce a positive Bayesian balance between its assumption and the data it accounts for?

So the question arises, what is the data we have to test theories of consciousness?

And there's a big problem, which is well known, that consciousness is a private experience.

And so there's not much public data about it, but there are several kinds of data we can use, such as data about levels of consciousness, alertness and being asleep and so on.

Data about quality and they are problematic because they're private phenomena.

But nevertheless, you can two people can agree something about similarity scales of their quality.

Like different colors being similar or different.

And the form C is the shapes of things in spatial consciousness.

And I shall argue that the main manifestation of consciousness, which we have, what we're experiencing right now, is a 3D spatial model of the world around us.

And that 3D model is very detailed and quite faithful to the real world around us.

And so that has a large information content.

Going down the list, there's conscious imagination.

There is consciousness of the self.

There are anomalies and illusions of consciousness.

And there is the hard problem, if you like, the existence of consciousness itself.

Now, I shall argue that point C, the shape of spatial consciousness, gives us much more data to test any theory than any other type of data.

So what's the testable data in spatial consciousness?

Spatial consciousness is like a 3D model of the real world around you.

That's what you're experiencing right now.

And it has a lot of the geometry of the real world.

And particularly straight lines in consciousness correspond to straight lines in the real world.

And you can test this.

I won't go into the details of how.

So this agreement between our conscious experience, which is quite information rich, and the real world around us, which is obviously information rich, gives a lot of data to test any theory of consciousness.

And I suggest roughly it gives thousands of bits, even in a few seconds.

And I shall suggest that no other form of data about consciousness comes anywhere close to this in terms of testable information content.

So if you really want to prove your theories right, then it's got to account for the spatial form of consciousness if you're going to believe it.

And down at the right here, I put my contention that spatial consciousness is, if you like, the elephant in the room consciousness theory.

So we have to consider how the brain does spatial cognition.

And all animal brains pretty much do that.

They have to understand 3D local space for a very fundamental reason.

Just to control their every movement, to move every movement of their limbs.

They have to understand 3D space around them.

And there's been huge selection pressure to do this well.

And we believe that animals do it very well.

Even insects do it quite well.

And also, it appears that most of the animal brain is devoted to that.

Now, if we assume a Bayesian hypothesis about cognition, then the way animals do that is to build a maximum likelihood 3D model of local space in their brains from their sense data.

So it becomes fairly natural to expect that spatial consciousness is going to be linked to this spatial model.

And the question is, is it a purely neural model, as in classical neural models of the brain?

And if it is, how does neuron firing represent 3D space?

And this is quite a difficult problem.

And really, there are very few viable solutions to it.

But I shall switch now.

I shall put that problem on one side and switch to the projective consciousness model, PCM, which has been developed by David and his collaborators since about 2017.

And it's built on the free energy principle.

So it's based on a Bayesian model of local space in the brain.

And for neural process model, it seems the process model of the free energy principle.

And so the PCM assumes that the brain holds a 3D projective transform model of local Euclidean space.

And the key thing is that PCM then predicts that spatial consciousness is like that model.

And this is a very good prediction.

It agrees very well with data.

For instance, the trivial fact that straight lines in consciousness are straight lines in the real world.

So the PCM has a good Bayesian balance.

I should say this is very distinctive amongst many theories of consciousness.

Very few of them talk about the spatial nature of consciousness.

And if they do, they probably treat it as a bit of an afterthought.

But the PCM puts it out in the front.

And I think that's a very positive feature of it.

And it has a good Bayesian balance.

And it does make various other predictions, which people have talked about in the papers.

Makes predictions about illusions of consciousness.

And makes predictions about predictive imagination and planning actions.

Now, I don't think those are quite as significant in the Bayesian balance, in the agreement with the data, as the basic agreement with the form of consciousness right now.

Consciousness from sense data.

But I shall say the PCM has three difficulties.

I shall emphasize they're not difficulties of the PCM alone.

I believe they are difficulties of any neural theory of consciousness.

And I'll just list them briefly here because there's going to be a slide describing them.

The first difficulty is neural selection.

How do we pick out just those neurons which cause consciousness from all the others that don't?

And the second one is neural precision and speed.

We know that our conscious model of space is a very detailed and precise one.

Can you do that with neuron firing?

I would suggest you can't.

And the third one is a complex decoding problem that, anyway, I'll come on to that.

So I'll describe these three in a minute.

But the emphasis is that they are not just problems of the PCM.

They are difficulties for any neural theory of consciousness.

For instance, IIT or global neural workspace theory and so on.

So what are the three difficulties?

The neural selection problem.

Most neurons in the brain are not linked to consciousness.

We know that from cortical lesions which don't interrupt consciousness or the cerebellum or Bjorn Mirko has written quite extensively about this.

But neuron theories of consciousness say that the activity of some neurons causes consciousness and not

others.

So the question is, how are those special neurons selected?

And as far as I know, there's no sound physical basis to pick out some neurons from all the others.

So you build in some selection assumptions into your hypothesis.

But those form a debit in your Bayesian balance.

So they're a debit for any neural theory of consciousness.

The second difficulty.

Whoops.

All right.

Second difficulty is how do neurons represent space?

Now, there are several possible ways of having neurons representing 3D space.

You can represent 2D space, all right.

Can you return to sharing the screen?

Oh, sorry.

I've lost sharing, have I?

All good.

So how do neurons represent 3D space?

There are several ways they might do it, but they all have problems of precision and speed.

For instance, there is a Bayesian computational model of how animals build 3D internal model of space from their sense data.

And this is already on one of the All live streams at this YouTube address.

But that model shows that in order to build a 3D model of space, even for a small animal like a bee, you need memory storage errors to be roughly the order of 1%, as small as that.

Whereas if you model neurons accurately, properly, they don't give you this precision.

Because in a neuron, most theoretical models of neurons are ideal abstract neurons whose outputs are real variables of high precision.

And that's shown in the top of this picture here.

Whereas actually neurons outputs are spike trains, which in the short timescale gives very limited precision.

For instance, in a fifth or the second possible timescale for consciousness, you may get about 10 pulses.

And that gives you a precision of only 30% in that timescale.

So the precision you get from neural representations is very different from what you need to build a decent 3D model or what we experience in consciousness.

And this, I think, hasn't been really spotted very much because most of the neural models that you see in the literature, they don't model spike trains.

It's difficult enough to model networks of neurons.

And so they use abstract, precise neurons.

So that's the second difficulty.

How can neurons store a precise and responsive, fast model of 3D space like the one we experience?

Finally, can you see that change?

Third difficulty.

Third difficulty is what I call the hard triangle.

Because conscious experience, which is C in this picture here, is a faithful, hardly distorted model of the real world.

There's the real world, A, and there's consciousness awareness, C.

But what the model we have found in the brain are two-dimensional models and very distorted models.

So there are many different models and they're highly complex encoding to get from A to B.

So if you want to get back from B to C, you have to do very complex decoding to undo all the complex coding that most species brains do.

And so ABC is a hard triangle.

And so ABC is a hard triangle.

You have to have complex decoding assumptions in any neural theory of consciousness.

And the worst thing of that maybe is that the decoding has to be species specific.

It can't be universal decoding.

So those are the three difficulties for any neural theory of consciousness, including the PCM, I believe.

So that's where we've got to.

I've covered topics one and two of this talk.

And now I think we'll break for any discussion or contributions.

Thank you very much, Robert, for your presentation up to now.

I can follow it very well.

I think you point out very important issues.

I should say, and perhaps, you know, we scan that from the standpoint of the PCM, even though indeed we have framed it in the context of active inference and more specifically the free energy principle, we do not necessarily.

It does not necessarily entail that we buy or we endorse or we believe in the current neural network models that are used in this context.

Personally, I've been a neuroscientist for many years.

I've done a lot of electrophysiology, lesion work with human patients.

Also, I worked with rodents.

I've done a lot of functional imaging and modeling.

And at some point in my career, I thought that I did not understand the brain well enough.

And I stopped doing neuroscience.

I stopped caring, quote unquote, about the brain.

And for us, what is the most important in the PCM at this point is all you said, is that we can capture some aspect of this phenomenology that seems essential, etc., etc.

So personally, I have given up at this point trying to devise or figure out any, you know, realistic neural network models.

And when you take just one neuron, as you said, Robert, it's a universe.

You just think about how receptors are up or down regulated through fast gene expression, how dendritic

trees and their complexities and their geometry plays a role in computation.

I don't believe that the spiking of the neurons, which we often call the output of the neuron, are necessarily where we should be looking at.

And I don't know where we should be looking at.

I committed an odd paper a long time ago in 2015, in which I kind of said, well, I explained why I had given up hope at this point about brain science to account for that.

Because I think we don't understand the brain.

I think there are principles we haven't figured out yet by far.

And my guess was that a bit like by analogy with computer science, there are layers of abstraction, mechanism of virtualization that hide the detail of the hardware, which we don't understand at all.

And so, so far, it's a very open-ended thing.

So to me, what matters the most about the PCM at this point is with respect to the brain, it is neutral.

We don't know.

At this point, I have absolutely no hypothesis, except very gross anatomy hypothesis, but are some, you know, possible details about, you know, but you're right.

So, I think that's the problem, but that's the hard problem.

And personally, I have given up on trying to account for it.

Thanks, David.

I find so much to agree with there.

And so, yeah, we don't understand what's going on in the brain.

Maybe neurons are doing something much more complicated than simple spike trains, but maybe there's also this wave.

And we'll talk about that.

Carl, Ken, do you have any comments on this stage?

Carl, you want to go?

Or I can, if you don't mind, I'll just follow up with, I completely agree with what David said.

And I will say my own sort of approach to the problem has always been sort of agnostic about the underlying realization.

So, I've kind of started from, you know, in my own thinking anyway, always from the point of view of phenomenology with some kind of, you might say, general acceptance of some sort of physicalism.

But agnostic about the details of that physicalism.

So, agnostic about whether it's exactly what goes on in the brain or what level and so on.

I guess I've tended to, even though I agree you can't exclude it a priori, I've tended to kind of think, well, quantum mechanical theories are probably barking up the wrong tree.

But that's just like my gut on that.

And then I...

I'm sorry, personally, Ken, at this point, I'm so confused about the brain and about the physical nature of the universe that I'm even totally open that the whole thing is quantum.

I have no idea.

Okay.

Yeah.

Yeah.

As I said, it's just sort of like my gut on that, but not...

I can't really defend it in detail for a number of reasons owing to my own ignorance.

But...

And I would just add that...

And I agree.

I think the way you presented the PCM generally was exactly right.

You did mention that we do have something to say about imagination, like action planning, and about anomalies and illusions.

And we agree with you, I think, that the main set of data we aim to capture are the spatial component of consciousness and the perspectival spatial component, like what we use projective geometry to capture.

But then we do try to, as you know, try to derive certain other bits of parts of the relevant data from that basis.

So accounting for illusions, like the moon illusion and the Ames room illusion, and accounting for imaginative action planning.

And also, I should add some data about self-awareness.

So we have a paper on self-awareness in this context as well.

So, yeah, I think that's all I got so far.

But this has been very interesting so far.

Great.

Thanks, Ian.

Carl?

Carl, can I make any comments?

No.

I'm just enjoying this as a spectator, batting for both sides.

One comment, and it seems to me that we're focusing on the distinction between principles about how the brain makes sense of its world and the process theories that attend those principles in terms of instantiation.

And there's an agnostic approach I think you can take in relation to process theories.

So I'm interested to see how much Robert's focus will be on the process theory and the actual substrates, which I think is interesting from the point of view of actually simulating this kind of inference and also being able to reproduce in silico in terms of creating a kind of artificial intelligence that has this kind of competence.

So there may be a more pragmatic focus.

I was intrigued by the previous slide, this hard triangle.

Where are you?

Yeah.

Just go to it.

There.

Yeah.

This, yeah.

I didn't quite resonate with this.

I think for a start, the notions of encoding and decoding for people like me should be usefully unpacked.

um so everything in terms of belief updating is some kind of mapping or morphism or relationship

um so i'm assuming that thing is a good word for it yeah okay so the world generates sensory inputs

and then the brain um represents those inputs i mean for b you just imagine you know the visual

cortex with a highly distorted version of space in it all around the fovea and so on and all the other

cortical maps are highly distorted representations of space somehow you have to cram all those back

into an undistorted representation so you've got to decode the encoding that's gone on

i see so right so the conscious awareness is still part of the brain somewhere

it must come from the brain somehow i think yes then we all i see so you're using encoding and

decoding in a sort of um uh distorting undistorting it's like a distorting mirror you've got undistorted

but i guess it relates to carl distinction between neural neural substrate and process theory because

indeed if you look at the classical sort of quote-unquote 2d maps in the cerebral cortex

uh visual cortex with your retinotopy where you see a lot of deformation but also somatosensory cortex

you think about the classical you know map of the body uh uh and you are saying well if that's the way

it is kind of represented quote-unquote at the level of that substrate then since it is deformed and we

don't experience it that way we should somehow have a process on top of that that use some morphism to

transform this different i'm saying our theory of consciousness has to tell us how that how about

unscrambling distortion happens yeah but i believe that the level of observation that we are considering

here for instance those 2d quote-unquote 2d because they are not really 2d but uh cortical maps uh

retinotopic or somatotopic or you know topic maps uh are very quote-unquote low level of processing

uh which might be very far away from yeah what is going on there might be some other super representation

somewhere but we've not found yet yeah and it might not be visible at the level of the anatomy

well yeah those are alternatives i agree okay so sorry carl i interrupted but i wanted to make

this precision too yeah no that clarifies it thank you but carl i mean i think we can all be agnostic

about certain things in the process model i think you and i at least are not agnostic about basism

them yeah i i would put sort of you know the notion of um sentience of the kind we're talking about

as a process of basing belief updating i i put that at the sort of principle level so at the mechanics

that that doesn't commit to um an infrastructure or a realization or a process that that realizes that

um you know that kind of principle or that kind of uh mechanics um so i think the spatial wave

theory is really a commitment to a process theory um that could you know happily accommodate basing

belief updating and its particular application the context of spatial perception and projective you

know the projective consciousness model

is

Good.

Okay, so moving on.

So we've done one and two, and what we're going to do in three is add something to the PCM.

And the thing we're going to add is not a little thing.

It's the assumption, the hypothesis, that neurons don't do everything, but there is a wave in the brain which

stores 3D spatial information.

So that's what I'm going to talk about next.

So here's the hypothesis.

There is a wave in the brain that represents 3D space.

So it assumes there is some nearly spherical region of the brain with a diameter D , and it holds waves, which couples the neurons.

And there the sphere is illustrated, and a couple of waves, wavelength λ_2 , that green wave, and wavelength λ_1 , a shorter wavelength, that red wave.

And those waves represent things in space, and they represent the red thing, R_1 , and the green thing, R_2 .

So there's a kind of inverse relationship there.

The shorter the wavelength, the longer the distance to the thing that's being represented.

But obviously, if it's a spherical region, you can have waves in any direction.

So you can represent any direction and any distance up to a maximum, which is defined by the minimum wavelength.

So that wave can represent lots of different things.

And the hypothesis is that that wave acts as a working memory for space.

It's really like a hologram.

Now, one issue with that is that if you want to represent objects at very large distances away, then you get into wavelengths that are too short.

So that's why you may need a projective transformer of space to pull in those long distances to be able to represent them on the wave.

By the way, the evidence for this wave is going to be described in another YouTube live stream, which we're making next week.

It's going to be at that address.

So how well does a wave-based spatial memory form?

It can represent approximately D over λ cubed independent positions, which is quite a lot.

And it can give you spatial precision, the order of λ over D .

And the point is that this can easily be 1% or better, which is the precision you need.

So there are significant benefits to wave storage.

It has high capacity.

You can put many objects in it.

It has high precision.

It can be fast.

And it has low spatial distortion.

It is just like a hologram.

And it's a very simple and direct representation of space.

So a spoiler looking ahead to part four of the talk is that this is very much like spatial consciousness.

Question, is there any evidence for such a wave?

I think there is evidence.

And there's three kinds of evidence.

First two comes from brain anatomy, because there are spherical regions in the brain which are very well suited to hold this wave.

In the mammals, there is the thalamus.

And that's shown in green on the left of the picture.

And you see it's approximately spherical.

And that is very suitably connected for holding spatial information.

It's connected to all the sense data.

And it's connected to all the bits of cortex that work out what to do with the sense data.

And similarly, in the insect brain, there is a central body in the very center of it.

That is approximately spherical.

And it's very well preserved shape in all insect species.

So those are both pieces of evidence for a wave in the brain.

There is a place for it.

So I believe there are three lines of evidence for a wave.

One is the computational neuroscience, which I've mentioned before, that spatial cognition is a very important thing.

But neural memory for it is too imprecise and too slow, whereas a wave storage model can possibly overcome these problems.

So that's the yellow evidence.

The evidence from the insect central body that is nearly round, very central, very well connected.

And there's evidence from the mammalian thalamus.

It's also round and well connected.

But there's also what I call a smoking gun for a wave in the thalamus.

If you don't assume a wave, the anatomy of the thalamus doesn't make sense.

But I won't go into that in this talk.

So together, I believe these give quite strong support for the wave hypothesis.

In fact, they give it a probability of 0.5 approximately, even if you try and weight the odds against it.

As I say, you can see the discussion of this in another YouTube live stream we're about to make.

So.

Then that's on from part three to part four of talk.

Assume there is a wave in the brain which holds the near projective transform of 3D space.

And this wave is serving a very essential role to understand space, to control all movements for any animal.

And the hypothesis simply is the wave is the origin of consciousness.

That neurons have nothing to do with consciousness at all.

It comes entirely from the wave.

This solves three problems, the three problems that I outlined before.

Firstly, the neural selection problem.

You've no selection problem because neurons have nothing to do with consciousness.

The precision and speed problem.

The wave has sufficient precision and speed to do it.

And the decoding problem.

The coding in the wave is just a Fourier transform.

So it's a very simple coding and encoding.

And it's a very simple decoding to give you back consciousness experience.

So it has solved those three problems of neural theories.

But the main thing is that spatial thought like the PCM, preserving the merit of the PCM, the spatial form of consciousness agrees very well with our conscious experience.

It has to be projected because we need the projected transform to pull in objects from very large distances to represent them by a minimum wavelength.

So in summary, the projected wave theory has a positive Bayesian balance.

Doesn't have any nasties about it.

And it agrees very well with the main source of data about consciousness.

So just some aspects of this theory.

What is the physical nature of the wave?

A big objection might be we've never seen the wave because I believe it's probably not an electromagnetic wave or electromagnetic field.

And the main reason I believe that is that electromagnetic fields on their own don't store information for the time periods we need to store information.

And now electromagnetic fields might be part of the answer, but I think they're not the whole answer.

And the big question then is why have we not detected this wave in the brain?

It's been there all these years.

I think the reason for that is we will expect if there is a wave representing space, we will expect it to evolve, to have very low intensity.

So its energy consumption is very small.

And having that low intensity, we simply haven't known how to look for it.

Now, coming to quantum possibilities, the wave might be a coherent quantum excitation.

And there's increasing evidence that quantum effects do play a role in biology.

And it might even be a boson condensate, a BEC.

Now, those examples of those are superconductors and superfluids.

They're a very rare state in nature.

But the key thing about a BEC is a superconductor can hold information for unlimited time.

So it would be good for memory.

And they're a very rare and distinctive state of matter.

And so if you assume that consciousness comes from them, you don't get into a trap of panpsychism.

Or it might not be a quantum effect.

Because I think that this question is entirely open.

So we'll set that on one side.

And that's future exploration.

There's an issue about the evolution of consciousness.

We know that consciousness is our consciousness is like the real world.

It's very like the real world.

And the question is, why is this?

And you think, well, it must have evolved that way because some need for it to be like the real world.

But in neural theories, there's a problem.

Because neural computation in the brain, the classical neural model, is classical dynamics, not quantum physics.

And it's local.

And so consciousness is a non-physical, non-local effect.

And they don't have any effect on neurons firing.

So the result is that consciousness in a neural theory has no causal effect on fitness.

That means there's no selection pressure on it to make it the way it is.

So in a neural theory of consciousness, we're driven to this slightly awkward conclusion that the special form of consciousness that we have is just a bit of an accident.

That seems very unlikely.

Whereas in the projective wave theory, there is a good evolutionary reason for consciousness being like it is.

That is, that 3D spatial cognition is very important.

Wave is an essential part of it.

So the wave has been shaped by evolution.

And it's been shaped to be just like the real world because that's the best way of controlling movement.

So consciousness then is determined by the wave.

And that means consciousness has evolved to be like the world.

So the wave theory solves another problem there.

Finally, the unity of consciousness.

You may ask this question occasionally.

Why do I only have one consciousness and not several parallel consciousnesses?

And this also is something of a difficulty for neural theories because we have plenty of neurons in the brain to have many parallel consciousnesses.

And brains can do some things in massive parallelism.

And if we had multiple parallel consciousness, it might be useful, for instance, for parallel planning of action to see which one works.

That's a difficulty for neural theories.

The wave theory doesn't have this difficulty because it accounts for the unity of consciousness.

There's only one wave in the brain.

That is the source of consciousness.

So there can only be one consciousness.

I think I'm on to the last slide now, pretty much the last slide.

If this theory is taken seriously, it makes consciousness into a scientific problem rather than a philosophical problem.

Because a wave in the brain, if a wave in the brain is the key to understanding consciousness, well, if there's no wave, then this theory is wrong.

Get about it.

If there is a wave, I think there are very strong reasons to link it to consciousness because it has such a good fit to consciousness.

So whether or not the wave exists, that's a scientific question.

It's not a question for philosophy.

And I think there's a good prospect of answering it in the next 10 years.

So in 10 years' time, either this theory will be the only theory of consciousness or it will be wrong and forgotten.

I think that's a virtue in a theory.

So finally, last slide on the hard problem, David Chalmers' hard problem.

The projected wave theory doesn't solve that problem.

What I would say about that is all science has hard problems.

There is the hard problem of consciousness.

Why is consciousness linked to matter?

There are hard problems in physics, like what is space-time or what is the most fundamental particle?

Or the hard problem of cosmology.

Why is there a universe?

So hard problems just don't have an answer.

Scientists, we're like children, keeping asking why.

Say, why this?

Why that?

And we can go on asking why for as long as we like.

So for instance, for fundamental particles, we thought there was matter.

We thought there were atoms.

We got to protons.

We got to quarks.

And we keep going on.

And we'll never get the final answer.

But we still do science.

So I'm not too worried about the hard problem.

Consciousness.

So that is all.

And I've just laid out a few questions we might want to discuss in the remaining time.

Maybe you could share your slide again so we can see the question if you have them on the slide.

Sorry, is it not sharing now?

I see the slide still.

Yeah.

So you can leave it.

Maybe check your view, David.

And then you can leave that up and we can have a panel discussion.

Okay.

Well, thanks a lot.

Those who are here, Ken and Carl and David,

please feel free to comment.

And also people will write some questions in the chat and I'll read one in if you pause.

But anyone here who would like to give a reflection or any comments, that would be awesome.

Could I go ahead?

Can you hear me?

Yeah.

So if I may, I basically agree with what you say about the hard problem.

Just to take the very last thing that you mentioned about the hard problem.

And this is why earlier on when you said that you had kind of a list of sort of, you know,

when you're the idea of the Bayesian balance for theories of consciousness, and you mentioned the existence of consciousness as itself,

something that might be a problem for all neuronal theories of consciousness.

I think you did that in the slide too.

Any theory, yeah.

Yeah.

So I agree.

I think it's a problem for any theory of consciousness, even the kind of more metaphysical theories that people who reflect,

like Chalmers and others, panpsychists, who reflect on the hard problem a certain way, end up with.

You can actually raise the same set of questions for them.

So there's really no progress there, in my opinion.

So in other words, if you take the hard problem as a kind of indication that, well, consciousness couldn't be physical,

or either that or the physical world itself must just be consciousness, which is kind of the two directions they go there.

I think that's kind of misguided.

I mean, we have to sort of accept that our ability to penetrate, you know, cognitively a priori into the nature of the cosmos is limited.

So the best we can do is make models and, you know, based on the phenomenology, and try to arrive at a model that would give us some kind of neural or physical, to speak more broadly, physical testable, you know, hypotheses we can test about the physical world and what we think realizes consciousness.

And it seems to me like that's exactly what you're doing, and that's what I would think is right, too.

But I did want to ask you, when you say that no neurons are linked to consciousness in your own model, do you mean, maybe I just don't understand exactly, but...

They're not linked directly to consciousness.

But they're somehow involved in the realization of the wave.

Oh, they are. Absolutely.

They're controlling the wave.

They're setting up the wave.

They're all on the cortex.

They're using sensitive data to fix the wave, and it's the wave that makes consciousness.

Okay.

And on the issue of unity, well, so suppose we identify this type of wave.

I mean, it's at least, you know, conceivable that we would find multiple such waves.

Would you conclude...

It is conceivable.

Well, I think the evidence in the thalamus and the central body is one wave.

Okay.

And I think one wave is what you need to understand one space around you.

Okay.

So, but if you found multiple such waves...

Oh, yeah, absolutely.

That would be an issue.

Yeah, okay.

I mean, would you conclude that there must be multiple consciousnesses then in a single brain, or would you think that the theory is wrong?

No, I would have a selection problem.

Just like neuron theories have a selection problem selecting neurons, I'd have a problem selecting which wave was conscious.

I see.

Rather than conclude that there are multiple...

You want to avoid that.

You want to avoid that.

And I...

My experience is I don't have multiple consciences.

Yes.

Right.

My data.

Right.

But that might just be because only one of them gets to report.

Yeah, absolutely.

Okay.

You know, like the way they...

You know, I know it's debatable, but the way they often interpreted the split brain cases coming out of...

Yeah.

Good question.

Good question.

I mean, I say one wave, but we have two thalami.

So the assumption there is that those waves are linked.

And perhaps in split brain, they don't get linked.

Hmm.

Okay.

I think two thalami have one wave in them, but that's a detail.

I see.

And then when it comes to the kind of wave, like...

So you don't think it's an electromagnetic wave.

You think it might be a Bose-Einstein condensate.

That's a hypothesis, I think, that Karen Smith once...

It's been around for some years.

Yeah.

Alexander Karen Smith in the book.

That's the...

Yeah.

Not the book on the origin, the famous book on the origin of life, but he wrote a book on consciousness back in the 90s, I think.

Yeah.

There was Michael Lockwood as well in the 80s, I think.

It's a long history.

Okay.

So how...

When you say that you think we'll be able to probe this within the next 10 years, I mean, could you sort of sketch how that might go?

How we might be able to...

Yeah.

I mean, just...

This is not...

This is probing the idea of the wave.

Right.

As I say, I think that there's three pieces of evidence.

There's neural models that you can't build decent neural models.

There is the thalamus and there is the insect central body.

I think we can go further on each one of those.

We can test them.

We can test the thalamus ideas much better using connectome data.

We can test the central body using connectome data.

And we can test the computational models by just trying to build them and see if we're getting it.

So there is science to be done to falsify this hypothesis.

And it may succeed in falsifying it.

Okay.

But how would you measure such condensate?

I mean, if you want to be able to demonstrate the existence and the property of...

Ideally, you'd like to actually directly detect it.

I think it's a bit like when Pauli in the 1930s predicted the neutrino.

He thought it was absolutely impossible to detect neutrinos.

And it took 25 years to find it.

And I think it will take a long time to really find this wave in the brain.

Because it's very low intensity, probably.

And because we just don't know how to look for it.

And I think we have to do the biophysics.

Until we have a convincing case for,

Ah, it's this kind of wave.

And then we can build a detector.

But until then, I think we won't detect it.

Like neutrino or like dark matter.

Similar thing predicted theoretically years ago, but still not detected.

I'll read a question from the live chat.

Okay.

Andrea Val wrote,

Thanks for sharing such interesting ideas.

I wonder how the PCM or 3D model can explain or deal with the un-consciousness of emotions.

So where is affect and such in this kind of spatialized model?

Yeah.

You want to do it in the PCM first, David?

If it's okay.

Robert, maybe you have read some of our recent work on that.

But I have a specific answer to that question.

Yeah, you go first.

Yeah, sorry about that.

Well, one of the things that we realized working with that particular geometry

is it transforms information in the sense of information in space.

You can think about distribution in space, for instance,
that are going to be changed by projective transformation
in a manner that affects directly entropy.

So if something is very far, you know, and you have some uncertainty about the localization
of these things, so there is a probability function that represents that uncertainty
about the localization of that thing.

And there is only one thing in your field of consciousness to simplify the problem.

Then the entropy of the whole space, if you consider the whole 3D space
and some scalar on it that would represent probabilities, is very small.

Whereas when you get closer, for instance, by imagination,
so projective transformation that anticipates action,
you get the object closer and you transform also the information,
the variance, if you wish, of your distribution, so it becomes broader.

So the entropy increase, actually, as an effect of projective transformation
on the distribution of information.

So how does that relate to emotion?

What we realized is that it can account for some drives that are related to exploration
or exploratory behavior or search behavior to reduce uncertainty or maximize information.
transformation.

And that just does that.

So basically, the projective transformation, by in virtue of making probabilities function
in space change their shape as a function of perspective taking, if you wish,
entail or induce a drive to approach or avoid things or in this context to approach behavior,
either through imagination or through actual behavior, because that will maximize information
by increasing entropy of the object and you have more details.

So that's something that we wrote a paper that is currently under review in biological cybernetics
about that with Grégoire Sergent-Pertuis as first author, who is a bright mathematician.

And we can do a demonstration and actually derive theorems about that.

And also simulations that show interesting behaviors that depend on projective parameters

that will show that the behavior of the agent, depending on the parameter, will be more focused.

So they will tend to always approach one object because they are driven by this anticipation of increasing information through projective transformation.

But with other parameters or values of the parameters, on the contrary, they start now oscillating between objects.

They start losing focus as if they were kind of ADHD.

So there is a deep link between, let's say, the evaluation of space, the appraisal of space, the drives that unfold from this information and the behaviors that you have, for instance, approaching, and that geometry, that very geometry we are talking about yield, that type of thing.

So here we are talking about what is typically described as epistemic emotion, basically curiosity.

So we have natural curiosity and part of it that drives behaviors is a result of the very geometry that represents information in the place, according to that theory.

So I cannot enter into the detail on that, but you can also look on papers that we wrote in, what was that, 2022,

in the Journal of Theoretical Biology, in which we use little robots to demonstrate adaptive and maladaptive behaviors

with principles also derived from the geometry and the free energy principle, or something equivalent, to get a wealth of affective behavior, of approach avoidance, but also social cognition.

You know, you kind of, we can kind of simulate social anxiety.

We can, we obtain joint attention behavior as a result of mutual perspective taking.

And there are a lot of things we could account, that we can use in this model and in this framework to account for emotion, there is a lot of work ahead, but we believe that there is, and we have good reason to believe there is a deep link between space and emotion.

That's actually called aesthetics, that consciousness is a form of aesthetics, where space and emotion are untangled, and that we can fairly well account for many phenomena that relate to emotion and behavior through this model.

But that's, sorry for the long talk.

That's great.

Should I give my, my sort of answer?

It's very much along the same lines.

Emotion and space are deeply linked.

In the wave model, I mean, the project, the model of space contains all of space, the external space and the internal space of the body.

And an important part of emotions is feelings in the body, and they are part of your conscious awareness, part of the wave model or projective model, just like the others.

And so basically, this, the PW theory, and I think the PCM says, all consciousness is spatial.

We have qualia, but they exist in space.

You hear words, you experience colors, you experience all of those, and you experience feelings, all of them in space.

So if you look at all the components of emotion, human emotion, for instance, the words we say to ourselves and the feelings we have in our bodies, they are all represented in that conscious space.

So human emotion is a complex story, which perhaps we don't want to go into, but basically it comes down to the way we experience it is in that model of space.

And if I may, I would have that if you take the standpoint of pure phenomenology and you ask yourself, when I experience an emotion, for instance, deep anxiety or fear, something very intense that is very salient,

you can ask yourself how much of that phenomenology of that experience is actually, quote unquote, reducible to some spatial experience.

Things, for instance, you know, the experience of the body with the heart that starts pounding in your chest and you feel it stronger, your lungs that change there, you know, you breathe faster, for instance, and you can feel that movement in your body and all of that is spatial.

What you feel in your gut or your intestine gets kind of stretched or whatever.

We have metaphors, but actually a lot of the details of the phenomenology of emotion, of feelings, let's say, I believe is actually spatial, but it's overlooked.

People don't think about it.

But in particular, when you think about bodily awareness, there is a lot of spatial stuff going on here.

Yeah, I agree.

Mindfulness and meditation talk a lot about space and talk a lot about the body.

I think they put emotion in the right place.

Carl or Ken want to add anything, or I can look for another question in the live chat.

Well, I have one question.

So, I mean, your current hypothesis is that this wave is realized in the thalamus.

Is that right?

That's right, yeah.

Okay, so that's sort of, would, you know, there's this kind of debate about whether or not consciousness is cortical or subcortical.

So, this would definitely be on the subcortical.

Yeah, I mean, I think there's a lot of sense in the global workspace theory, and they were talking originally about a workspace involving the thalamus and the cortex.

I think it's, the consciousness really comes from the thalamus integrating all that spatial information.

Yeah.

So, it is down.

So, when, the, the follow-up, yeah, that's fine with me, frankly.

But, the, the follow-up question is, well, it's not exactly a follow-up to that.

It's more a general question about your way of thinking about it.

How, I mean, I know you, maybe you can't really answer this until we have a more specific hypothesis about exactly what kind of wave it is, but could you say something generally about how, the wave integrates with, or, or what's, what's the right way to put it?

Use, recruits, and, you know, the kind of computational apparatus of the brain, so that, you know, you end up, you decide to, oh, I'm going to reach out, you know, yay far and grab this, this bottle.

How does the, how does, you know, computationally speaking, how does the wave integrate?

Yeah, I think that's a very good question and could have a very long answer.

But, basically, the wave integrates information from all the sensitive data.

So, from your hands, from your ears, from your eyes, all that information goes through the cortex, and cortex decides where it's coming from

and broadcast signals back to the thalamus.

And the thalamus wave

integrates all those signals together

to produce

the Bayesian best model

of what's happening at this moment

based on touch,

proprioception,

hearing,

sound,

hearing,

vision,

and so on.

So, the computation,

it's an integration

of many sorts of information,

possibly in a cycle

between the thalamus and the cortex.

Okay,

but if,

if you need to,

you know,

for example,

program an action,

so you,

that's kind of the,

you've got your sensory,

the integration

of all your sensory information,

now you need to program an action.

How do,

and maybe this is just my ignorance,

how do waves,

how can waves be used

in a way to

generate,

program an action,

and then give it back

to your neurons?

I mean,

I've only got beginnings

of answers to that,

but basically,

a wave is very good

for representing 3D geometry

and for calculating displacements.

I know where my hand is,

I know where my coffee cup is,

so the wave helped me

calculate the displacement,

which way I have to move my hand

to pick up a coffee cup.

So it's that kind of

spatial calculation

that the wave is always

helping us do,

just to move our limbs.

I have a follow-up question

on that,

if I may.

Ken,

do you mind?

Yeah,

no,

go ahead,

go ahead.

You invoked earlier

Fourier transform

and inverse Fourier transform,

so at least 3D Fourier transform

and inverse Fourier transform

in order to do the encoding

and decoding

to, you know,

move from the wave

to the neural substrate

back and forth.

Of course,

as you know,

there is a limit

in resolution

when you do a Fourier transform.

you have the uncertainty principle

that rules

in linear spatial resolution

that we can have.

Of course,

it depends on the number

of samples.

I mean,

there are a lot of arithmetics

behind it that,

you know,

you can be more and more precise

as you have more and more

bits of encoding,

if you wish.

So you say that,

you know,

neural,

classical neural theory

is like build upon substrate

that lack enough precision

in spatial resolution,

for instance,

to account for our experience.

Would you think

that in the context

of a wave theory,

the type of uncertainty principle

that are attached to it

could cause a problem

of resolution too

or that it's the granularity?

There is a problem.

Yeah.

There is a problem
of resolution.

I alluded to it
in the slides.

That is that
if you've got
a minimum wavelength λ
and a diameter
of the sphere,
 d ,

the maximum precision
is λ over d
and that can't
be infinite precision.

So that's
the uncertainty principle.

But the idea
of the Fourier transform
is not that something
in the brain
does a Fourier transform.

It is the fundamental
bridging law
for consciousness
does the Fourier transform.

It says that
consciousness
is a Fourier transform
of the wave,
full stop.

And Fourier transform
is there
in fundamental laws
of nature
like quantum mechanics
for instance.

So it's saying
that this is a fundamental
law of nature
just like quantum mechanics
says momentum
is a Fourier transform
position.

So the bridging law
for consciousness
says the form
of consciousness
is a Fourier transform
of the wave.

I'll make a short comment
or Carl,
do you want to?

After you.

Okay.

Yeah,
just a few.

So thank you
for the presentation
and it's really
interesting on many levels.

First,
you and the paper
goes into more detail
on this.

You lay out
what your priors are
and kind of the
dependency structure
to actually reach
at least an estimate
of probabilities,
which is,
it's really interesting.

It gives a lot of

opportunities for people
to dig in
and say,
yeah,
I think that's 10%
likely or no,
it's a zero.
This is not possible
and kind of close
off a branch of inquiry.
And also this discussion
here about
the algorithmic
and the substrate
realization,
interaction
with the substrate
and even this
polycomputation
idea of the
observer's model
of what's happening
in a process
and kind of like
the dark matter there.
Like the observer
might suggest
that there's
a cryptographic process
though they couldn't
decode it
and then that would
have an entropy
distribution
that wouldn't
be distinguishable
in the absence
of the decryption

from a noise
but it would be
like still a hypothesis
kept alive
and you could say
I think it's a 1% chance
of this being the case.

One thought is
how high precision
really is
the spatial
generative model
in that there could
be visual upscaling?
We know there's
attention play
to make the periphery
look sharp
and have color vision.

Yeah.

And so it's possible
that especially
for like distant
objects,
small objects
that they could be
bundled
and just seen
as an assembly
as a larger object
if they're touching
like the car's
body and the wheel
and then
the same way
that in the periphery
color
and visual attention

seem to give
this extreme
like wow
that's like
exactly this angle
if you were to go
over there
with a protractor
but that could be
as we're sort of
seeing with diffusion
and generative AI models
that could be
quite a low resolution
structural
inference
followed by
a just
compatible
visual
realization
that also
and then
just last comment
really interesting
to explore
how does evolution
and intergenerational
change
shape
the substrate
that the sense
data wash over
so that non-stationary
stable states
whether they're
rate encoding
of neuron

whether glial
electro
and all these other
even potentially
more exotic options
how are those
stabilized?

Wow.

I mean
those are real
questions
to investigate
particularly
the precision
of our conscious
experience
and how much
is
quotes
an illusion
but I personally
think that if you
look at something
which is a straight line
you experience it
as a straight line
with some kind
of precision
that's of the order
of 1%
a deviation
of 1%
from a straight line
would show
and so
that's
a lower bound
on the precision

of your model
but again
there can be
lots of discussion
about this.

Yeah.

Interesting.

Carl?

Any
thoughts?

Well
just
a few
sort of
practical
and specific
questions.

So
the nature
of this wave
if I wanted
to
model it

I mean
are we talking
about a standing
wave?

Are we talking
about a soliton?

Are we talking
about spherical
harmonics
on the surface
of the
thalamus?

Are we talking
about
what kind

of basis
set might
I use
for this
kind of wave
even before
asking questions
about what
the wave
amplitude
actually means?

So what's
the functional
mathematical
form of this
wave?

very good
questions
whether it's
standing wave
or a moving
wave

I don't know
the nice
thing about
a boson
sign

condensate
is that

it acts
like a

long-lived
quantum
mechanical
wave function

in other
words
it has

phase
which can
be detected
and it
preserves
phase for
long times
and the
wave might
have to
preserve
phase
in order
to represent
positions
if you
have a
standing
wave
like
 $\cos$
 θ
in some
direction
in this
sphere
then there's
an ambiguity
it represents
plus
direction
and the
minus
direction
and what
happens
to the
minus

direction
so
I would
rather
have
phase
in the
wave
like
a
boson
sign
condensate
but
they're
very deep
questions
and I
don't
need
the
answers
I
mean
we
could
start
modeling
and see
what
happens
just
propose
some
answers
basically
you just
need

a
spherical
volume
that
holds
a
wave
excitation
with
a
certain
maximum
wave
vector
and
we
could
try
out
standing
waves
moving
waves
and so
on
what
would
differentiate
a
condensate
from
a
formulation
in
terms
of
a
time

dependent

scrolling

a

wave

equation

well

I

think

your

time

dependence

showing

a

wave

as

everybody

knows

the

problem

with

quantum

in

the

brain

is

that

decoherence

smashes

it

in

10

to

the

minus

20

seconds

but

that

doesn't
happen
to a
Bose
Einstein
condensate
a
long
lived
quantum
state
and
so
that's
an
ordinary
Schrodinger
equation
decoherence
is going to
smash you
in no
time at
all
unless you
have some
incredibly
clever
insulation
like in
a quantum
computer
right
I mean
lifting from
the
microscopic
to the

macroscopic
I mean
that would
suggest that
you need
very very
low temperatures
thermodynamically
speaking
yeah
nobody
has observed
a BC
at room
temperature
yet
but we've
got close
and the
idea is
that nature
might be
cleverer than
we are
one
angle
perhaps from
the quantum
FEP
from
Carl and
Chris Fields
and others
is
and so
what I
heard in
the

in the
basis
set
question
Carl
was
this
brain region
that hosts
the functionality
or a core
aspect of
the functionality
like one
option is
it is like
the wave
is represented
inside of
that sphere
like
just
there's probably
multiple ways
even that
could happen
but it's
like a little
reconstruction
wave
but symbolized
another option
would be
like a
holograph
the
representation
of space

is encoded
on the
interface
of that
region
with other
regions
and so
that kind
of
surface
is maintained
and stabilized
and has
information
flowing through
it
and then
there
wouldn't
necessarily
be
in the
middle
but it'd
be more
like a
ripple
on a
surface
which was
the
basis
set
question
Daniel
that is
very

interesting

because

in the

thalamus

there is

the

thalamic

reticular

nucleus

TRN

which is

a thin

skin

around

the

surface

of

the

thalamus

and

you

could

just

say

that

is

holding

the

surface

wave

and

the

wave

is

projected

inside

from

the

surface
it
would
be
useful
to
engage
Chris
Fields
here
there
are
two
very
distinct
uses
of the
word
holographic
here
you could
be
referring
to the
holographic
principle
whereby
all the
information
on the
bulk
on the
inside
say
of the
thalamus
was
on the

surface
on the
reticular
nucleus
of the
thalamus
for example
or just
on the
surface
in one
sense
and
that is
what the
rest of
the brain
sees
and possibly
consciousness
sees
as
that's
a very
different
use
of the
notion
of
a
holographic
projection
using
constructive
and destructive
interference
to encode
things

so
then you're
absolutely
right
my question
was really
searching
to distinguish
are we
talking about
a wave
function
in the
thalamus
so are we
talking
about
application
of the
holographic
principle
to
the
classical
information
on the
surface
of
some
bulk
that was
the
thalamus
and you
could then
argue
if you
wanted to

that
the bulk
which can
never be
observed
from the
outside
unless you
are the
thalamus
you never
know what's
happening
on the
inside
you could
only observe
what's
on the
surface
that would
be a
different
notion
of a
wave
theory
that would
certainly
be consistent
with quantum
information
theory
and also
consistent
with
Chrisfield's
inner screen

hypothesis
that could
be
spatialized
and endowed
with
a
in virtue
of the
fact
that
on the
holographic
screen
itself
on the
surface
there can
only be
classical
information
and that
information
can now
be
cast in
terms of
classical
Bayesian
inference
but it
also
relates to
Chrisfield's
inner screen
hypothesis
which is
you know

if you like
taking quantum
information
theoretic
principles
and trying
to apply
them to
consciousness
that eludes
the problem
of
applying
quantum
mechanics
that is only
relevant at the
very small
scale
or for very
precise or low
temperature
systems
you could
if you want
to apply
that kind
of maths
to
macroscopic
phenomena
then you have
to apply
it in a
scale
invariant
context
and I think

quantum
information
theory
would certainly
do that
so then
yes I was
exactly
sort of
trying to
probe
are we
is
the
is
PW
red
the
the
Bose
Einstein
condensate
variant
or alternative
version
of
the
projective
wave
is that
committed
to
quantum
mechanics
or is
it
really
an

expression
of
condensate
like
dynamics
in the
context
of
quantum
information
theory
and if
it is
does it
live
on a
holographic
screen
or does
it live
in the
bulk
and if
it lives
in the
bulk
what
are these
wave
functions
over
so these
are the
practical
questions
I was
trying to
get

very
interesting
questions
and it's
all to
play
forward
I
have
an
open
mind
about
them
but
my
feeling
is that
the
BEC
is
a way
of
getting
past
this
decoherence
argument
from
Tegmark
and others
that
neurons
have to
be
classical
and
the BEC

it does
work very
well as
a memory
that's
the
practical
point
of it
working
memory
I think
is what
you need
for
space
awesome
perhaps
each
can give
a last
rounder
thought
like
where
do they
feel
this
takes
them
or
what
are
they
looking
forward
to
and

Ken
then
David
and then
Carl
and then
Robert
I
yeah
I don't
have
anything
new to
add
just
that
I
find
this
really
interesting
and
I
remember
reading
Karen
Smith's
book
kind of
arguing
that
Bose
Einstein
condensates
might be
important
for
understanding

consciousness

and

thinking

that

among

the

quantum

mechanical

hypotheses

I like

that one

the best

so

I

embrace

but

as I

said

earlier

I just

sort of

have

this

kind

of

bias

but

I'm

totally

willing

to be

talked

out

of it

and

yeah

the reason

I mean

I don't
want to
go into
the details
but
I
think
Karen
Smith's
concern
was about
the unity
of
consciousness
primarily
and
also
kind of
made
the case
that
as you
were
suggesting
that
maybe
as you
put it
nature
is
cleverer
than we
are
and there's
some way
to realize
that
Bose

Einstein
condense
it
in the
brain
without
it
you know
the
decoherence
issue
but
anyway
I'm
just
to say
this is
all
very
fascinating
to me
and I
am
happy
to
continue
the
discussion
next
week
right
isn't
it
next
week
yeah
thank
you

Ken
and
for my
part
yes
also
looking
forward
to
continuations
of the
discussion
you know
my
direction
of
questioning
will
be
something
on the
lines
I
think
both
and
David
alluded
to
which
is
the
link
between
spatial
waves
in the
brain

and
the
projective
geometry
that
they
speak
to
and
the
information
geometry
that
underwrites
belief
updating
and
the
basic
mechanics
that
inherit
from
the
free
energy
principle
be it
classical
mechanics
classical
information
processing
or
looking
at
using
quantum

information
theory
to
read
belief
updating
I'm not
sure I'm
going to
get any
answers
but
that's
what I'd
be
interested
in
hearing
more
about
and
the
process
we haven't
talked about
mortal
computation
yet
so we
need
to
talk
about
mortal
computation
oh
yes
any

last
thoughts
well
the
thought
of a
man
who
is
50
now
and
feels
that
the
more
he
learns
the
less
he
knows
this
was
predicted
in
all
the
books
I
had
read
in
my
years
and
so
I'm

all
open
to
that
complex
universe
I
certainly
believe
that
nature
is
clever
than
I
am
for
sure
and
I
find
that
fascinating
I
like
the
rationale
it's
not
like
you
know
there
is
a
lot
of
stuff

about
quantum
mechanics
and
the
brain
and
consciousness

that
to
me
has
always
sounded
wishy-washy
if not
completely
wacko

but
I
came
back
a bit
from
that
I
had
a lot

of
like
can
priors
negative
priors
about
it
I

remember
once
in
Tucson
Arizona
at
the
famous
conference
about
the
science
of
consciousness
there
were
this
bunch
of
guys
who
called
themselves
they
were
German
the
quantum
consciousness
academy
and
they
were
all
dressed
in
black
and

they
are
all
you
know
headset
and
microphone
like
that
and
they
were
like
speaking
nonsense
as
like
a cult
but
I
found
that
very
fascinating
because
conference
back
in
the
day
when
it
was
very
psychedelic
but
that

was
kind
of
you
know
and
I
come
back
to
that
because
my
ignorance
I'm
becoming
aware
more and
more of
my
own
ignorance
I
said
well
why
not
and
definitely
I
think
that
maybe
Carl
can
try
to
do

that
if
we
could
get
Chris
field
in
the
next
goal
he
would
certainly
be
among
us
the
most
competent
I
tried
to
get
Grégoire
Sergent
Pertuis
to
the
call
but
he
was
not
available
he's
a
very

bright
mathematician
and
he
mastered
quantum
mechanics
like
if
it
was
you
know
it's
nothing
for
him
or
Daniel
Benquin
would
be
the
best
also
with
Grégoire
but
I'm
afraid
I
won't
be
able
to
have
them
here

but
I
think
it
would
be
very
useful
to
have
some
people
like
that
who
definitely
know
more
than
me
about
those
things
and
Chris
might
be
interested
to
join
I
don't
know
but
looking
forward
to
the

next
discussion
one
correction
they were
Swiss
David
they
weren't
they
weren't
they
weren't
they
were
Swiss
they
were
Swiss
they
were
men
in
black
I
think
yes
they
kind
of
look
like
people
from
the
matrix
like
they

were
Swiss
they
were
not
German
they
were
Swiss
and
they
had
they
had
a
leader
they
were
always
in
contact
with
wow
okay
yeah
please
Robert
and
and
and
we'll
contact
Chris
that's
been
really
fascinating
I

think
the
virtue
of
this
theory
is
that
it
is
falsifiable
in
other
words
if
there's
no
wave
in
the
brain
forget
it
this
theory
is
wrong
and
that's
exactly
the
question
we're
going
to
be
looking
at

next

week

I

look

forward

to

having

your

contributions

awesome

thank you

all

see you

at

16

UTC

on

July

29th

okay

thank you

thank you

bye

to

Thank you.